# Akanksha Sachan

## Contact

PhD Candidate in Computational Biology
Joint Carnegie Mellon–University of Pittsburgh PhD Program
**Machine Learning for Epigenetic reprogramming**
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## Education

**University of Pittsburgh School of Medicine** — *'24 – Present*
*PhD Candidate in Computational Biology* — *Expected '27*
Advised by Dr. Jishnu Das

**CMU, Kenneth C. Griffin School of Computer Science** — *'22 – '24*
*PhD Student in Computational Biology*
Computational Biology Department | GPA: 3.58/4

**Indian Institute of Technology (IIT), Bombay** — *'18 – '22*
*BTech. in Chemical Engineering* | GPA: 8.06/10

## Awards and Honors

**American Federation for Aging Research** (AFAR) Graduate Scholarship — *'25*
**Best Poster** Award at the Aging Institute Research Day, UPitt — *'24*
**1st Place, Student A100 track** | NeurIPS LLM Efficiency Challenge | team of 5 — *'23*
**1st Place, Accessibility Track** | Pitt Challenge Hackathon | team of 3 — *'22*
Recipient of **Undergraduate Research Award** (URA-01) by IIT Bombay — *'22*
Recipient of Merit-cum-Means (MCM) Scholarship by IIT Bombay — *'21*
**Top 0.2%** (Rank 3188 out of **1.1M+** candidates) in JEE (Main) — *'18*
**Top 1.5%** (Rank 2517 out of **160k+** candidates) in JEE (Advanced) — *'18*

## Invited Talks

**"Interpretable machine learning coupled to gene regulatory networks uncovers subcircuits underlying cell fate decisions"** | *preprint*

National Institutes of Health (NIH) IGVF Consortium Annual Meeting — *Sep '26*
International Society for Computational Biology (ISCB)/(ISMB) — *Jul '26, '25*
Keystone Symposia: Artificial Intelligence in Molecular Biology — *Sep '25*
American Association of Immunologists (AAI) Annual Meeting 2026 — *Apr '26*
Center of Lung Aging and Regeneration, School of Medicine, UPitt — *Nov '25*
30th Annual School of Medicine Graduate Student Research Symposium, UPitt — *Oct '25*

**"Sex-dimorphic DNA lesion accumulation drives a PDK4 metabolic checkpoint in skeletal muscle"** | *in-review at Nature Aging*

Center for Systems Immunology Annual Retreat, UPitt — *Nov '25*
23rd Annual Department of Immunology Scientific Retreat, UPitt — *Sep '25*
Aging Institute Research Day, UPitt — *Nov '24*

**"Micro-MPA digital twin, with sclerostin-dependent RANKL production reproduces pivotal dual-action effect of romosozumab found in clinical trials"**

50th European Calcified Tissue Society (ECTS) Congress — *Apr '23*

## Research Experience Prior to Thesis

**Mapping the epigenetic landscape of single-cells using 3-D chromatin architecture** *'22 – '24*

*Supervisor – Prof. Jian Ma, Computational Biology Department, SCS, CMU*

Developed a framework to identify cis-regulatory 3-D genomic hubs controlling target-gene expression and key transcription factors in healthy and cancerous mouse-brain single-cell, multi-omic profiling.

**Translational Mechanobiology Lab, MBI, National University of Singapore** *'22*

*Summer Internship | Supervisor – Prof. Hanry Yu, Dept. of Physiology, YLL School of Medicine*

Built a Generative-Adversarial-Network (GAN) pipeline for colorectal-cancer imaging to support clinical deep-learning deployment

**Lab for Bone Biomechanics, Institute of Biomechanics, ETH Zurich** *'21*

*Summer Internship | Supervisor – Prof. Ralph Muller, Dept. of Health Sci&Tech*

Added a Type-2 Diabetes pathway to an agent-based osteoporosis model and ran large-scale simulations on the CSCS Swiss National Supercomputing Center

**Multicellular Systems Engineering Lab, University of Notre Dame** *'20*

*Summer Internship | Supervisor – Prof. Jeremiah Zartman, Chemical & Biomolecular Engineering*

Built a morphometric feature-extraction pipeline for *Drosophila* wing imaginal discs (OpenCV, Fiji) to study organism-wide cell-signaling

## Key Coursework

**Machine Learning & Statistics:** Intermediary Statistics (CMU: 36-705), Probabilistic Graphical Models (CMU: 10-708), Introduction to Machine Learning (CMU: 10-701)

**Predicting genomic-hub mediated TF-perturbation response via geometric deep learning** *'23*

*10-708: Probabilistic Graphical Models*

Augmented a GNN for Perturb-seq response prediction with a Hi-C contact graph (Cancer cell-line), adding 3D-genome topology to the co-expression and Gene Ontology graph features.

**Single-cell DNA methylation prediction using single-cell HiC data and Graph NNs** *'23*

*02-710: Computational Genomics*

Developed a **GNN** to predict methylation from HiC data, connecting structure to phenotype; clustered **single-nucleus DNA methylation** data with K-means and t-SNE.

## Athletic Accolades

**Bronze Medal** | 35th Intercollegiate (inter-IIT) Aquatics Meet | 3rd in Medley-Relay *'19*

**3rd Position** | Comp-Bio Department, Pretty Good Race (5km) for SCS, CMU *'23*

**Swam 12-hours** | Swimathon organized by the IIT Bombay aquatics committee *'19*

**Ran 26.2 miles** | Dick's Sporting Goods Pittsburgh Marathon | 5 hours 15 mins *'24*

**Ran 13.1 miles** | Dick's Sporting Goods Pittsburgh Half Marathon | 2 hours 30 mins *'23*

## Leadership & Societal Contributions

**President** — *Jun '26 – Present*
*Allegheny Science Policy and Governance (ASPG)*
Previously Vice President (Jul '25 – Jun '26). Bridging the gap between **scientists and Pennsylvania legislators** through interactions at the **State Capitol**. Advocacy for **AI safety and biosecurity policy**.

**Technights Coordinator** — *Aug '23 – Aug '25*
*Carnegie Mellon University, Robotics Institute*
Organized and led a **computer-science outreach** program teaching middle-school girls sessions on foundation models, and deep learning.

**NIH Consortium Hackathon, Co-organizer & Team Lead** — *Mar '24*
*4D Nucleome, Seattle, Washington*
Led a team of 5 to build integrative pipelines for single-cell HiC analysis from cell-atlas-scale datasets

**Reviewer**
*ICML Foundation Model for Life Sciences Workshop 2026*, *PNAS Journal* (*assisted PI*), *Aging Cell Journal* (*assisted PI*)

**Teaching Assistant** — *Fall '23 – Spring '24*
*02-760: Lab Methods for Computational Biologists, SCS, CMU*
Taught a lecture on **genomics technologies** to a class of ∼20 students, revamping the RNA sequencing module of the course.

**Treasurer • Senator** — *Aug '23 – Aug '25*
*Graduate Student Association* (*Joint CMU-UPitt Comp-Bio PhD Program*)

**Editorial Board Member • Design Convenor** — *'19 – '21*
*Insight, the student newsletter of IIT Bombay*

**Convenor, Girls Swim Team** — *'19 – '20*
*Aquatics, Indian Institute of Technology Bombay*

## Consortiums & Workshops

**NIH Consortium Annual Meetings** — *'23, '25*
*IGVF, SenNet:Cellular Senescence Network, 4D Nucleome*

**Grace Hopper Celebration** — *Oct '24*
*AnitaB.org, Philadelphia*

**Computational Genomics Summer Institute** — *Jul '23*
*University of California, Los Angeles*

**Grad Cohort for Women** — *April '23*
*Computing Research Association* (*CRA-WP*), *San Francisco*

## Compute Allocation Grants

**ACCESS** (NSF Advanced Cyberinfrastructure Coordination Ecosystem: Services & Support) **1.5M** credits on the NCSA Delta GPU supercomputer — *'26*